Xiang Zhang

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Graduate Studies

Princeton University *2021-present*
Ph.D. Candidate in Economics
Expected Completion Date: May 2027

References

Professor Gene Grossman
Department of Economics
Princeton University
609-258-4032
grossman@princeton.edu

Professor Eduardo Morales
Department of Economics
Princeton University
609-937-5412
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Professor Stephen Redding
Department of Economics
Stanford University
reddings@stanford.edu

Professor Richard Rogerson
Department of Economics
Princeton University
609-258-4839
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Prior Education

University of Chicago *2017 - 2019*
M.A. in Computational Social Science (Econ track)

Renmin University of China *2012 - 2016*
B.S. in Mathematics and Applied Mathematics

Peking University *2014 - 2016*
B.A. in Economics

Fields

Macroeconomics, International Trade, Spatial Economics, Technology

Job Market Paper

Scale without Scope: Firm Specialization and Rising Industry Concentration (*JMP*)

Abstract: U.S. industries have become increasingly concentrated, with large firms accounting for a growing share of economic activity. Using U.S. manufacturing firm-product data from 1990 to 2020, I show that rising concentration is accompanied by increasing specialization among industry leaders. The average number of products produced by the top 1% of firms declines by more than 40%. I develop and estimate an oligopolistic model of multi-product firms, in which firms choose the product scope and invest in scalable, product-specific technology investments. As these technologies become more efficient, firms have stronger incentives to invest in their most productive products, while external competition and rising internal investment costs make peripheral products less attractive. Firms can therefore grow larger while producing fewer products. I structurally estimate the model using the U.S. firm micro data. Quantitatively, improvements in scalable technology efficiency generate increases in industry concentration and narrower product portfolios, consistent with the observed trends.

Working Papers

1. "The Economics of Agglomeration: Evidence from Grocery Stores" (with August Chen, Franklin Qian, and Qianyang Zhang).
   Revise and Resubmit at *Management Science*.

   Abstract: We provide the first large-scale causal estimates of demand-side agglomeration spillovers in the non-tradable service sector, leveraging data on 851 grocery store openings across the United States between 2019 and 2022. We combine a convolutional neural network with propensity score matching to identify credible counterfactual opening sites and compare business outcomes between actual and counterfactual locations. Grocery store openings increase foot traffic to nearby businesses within 0.1 miles by 23.4 percent within 6–12 months, with spillovers decaying sharply beyond this radius. The effects are strongest for wholesale and retail establishments and hospitality services, and are substantially attenuated after COVID-19—from 42.7 to 14.6 percent—consistent with a trip-chaining mechanism. Grocery anchors also reshape the local business landscape: within 0.1 miles, establishment growth rates rise by 7.0 percentage points and employment growth rates by 8.8 percentage points, averaged over the first three post-entry years. Decomposing establishment growth rates, the effect is accounted for by higher entry and fewer exits.

2. "Occupation Dynamics in General Equilibrium" (with Benny Kleinman, Ernest Liu, and Stephen Redding).

   Abstract: We develop a tractable quantitative model of automation and non-neutral technical change with dynamic labor mobility and capital accumulation. Given data on an economy somewhere along the transition path to an unobserved steady-state, we show how to solve for this unobserved steady-state and the counterfactual transition path in response to shocks. We invert the equilibrium conditions of the model to recover shocks that exactly rationalize the observed employment and wage data. We find relatively rapid convergence to steady-state, such that initial conditions are relatively unimportant in explaining the observed employment and wage dynamics. We find that intermediate-wage occupations have experienced the greatest automation, whereas high-wage occupations have witnessed the greatest productivity growth. We find that the combination of automation and productivity growth explains most of the observed employment and wage polarization in the data. We compute an inequality-output frontier, which summarizes the impact of automation and productivity growth on wage inequality and aggregate output, and can be used to assess the effects of emergent new technologies, such as artificial intelligence.

3. "Firm-to-firm relationship, uncertainty, and trade"

   Abstract: The recent surge in trade and industry policy uncertainty has led to growing disruptions in global value chain relationships, posing additional risks for exporters. I study the effects of relationship termination risk on exporter performance, investment, and search decisions. Event-study analysis highlights that the loss of major international buyers has a long-lasting detrimental impact on exporter performance. Firms losing major buyers experience a decline in export value and an increased likelihood of exiting the export market. These firms intensify their search efforts after the loss, yet their newly formed relationships tend to be short-lived. Inspired by these facts, I build an export dynamics model with search frictions and endogenous relationship-specific investment. Calibration results indicate that exporting firms face substantial search costs, as suggested by the existing literature. Moreover, relationship termination risk, and its subsequent influence on firms' endogenous choices, are primary factors explaining the observed firm dynamics.

4. "On the role of health in climbing the income ladder: Evidence from China" (with Gordon Liu and Franklin Qian).

Work in Progress

5. "The labor market dynamics of U.S. inventors" (with Meghana Gaur).

   Abstract: How do the labor market dynamics of inventors affect firm innovation? We provide new empirical evidence on how research productivity changes following inventor job moves, both for the moving inventors and for their former co-inventors who remain at the firms they leave. We link patent records to inventors' employment histories and use a matched event-study design to compare similar inventors moving to young firms and large incumbents. We find that inventors moving from startups to large incumbents experience a 47% increase in patent output. However, their departures are followed by a 17% decline in patenting among former co-inventors who remain at startups. We find no comparable decline among co-inventors following inventor departures from large incumbents. These findings reveal an asymmetry in inventor mobility: moving to large incumbents is associated with higher productivity for inventors but lower patenting among their former collaborators at startups. To quantify the aggregate implications of these effects, we develop a model of creative destruction with inventor mobility and examine how the allocation of inventors across firms shapes innovation and firm dynamics.

6. "Input Specificity, Capacity Constraint, and Supply Chain Resilience"

Conferences and Seminars

2025 NBER SI Micro and Macro Perspectives of the Aggregate Labor Market
2024 UEA European Meeting

Research Experience

2022-2023 Research Assistant for Professor Stephen Redding, Princeton University
2021-2022 Research Assistant for Professors Gregor Jarosch and Ezra Oberfield, Princeton University
2019-2021 Senior Research Specialist, International Economics Section, Princeton University. Assisted with research projects for Professors Gene Grossman, Eduardo Morales, and Esteban Rossi-Hansberg

Teaching

Princeton Macroeconomics - Fall 2026
Introduction to Macroeconomics - Spring 2026
Food and Agricultural Economics - Spring 2024, Spring 2025
Chicago Booth Microeconomics - Fall 2018

Honors Awards Scholarships and Fellowships

Dartmouth Globalization Fellowship	*Fall 2025*
Avinash K. Dixit Prize, International Economics Section, Princeton University *For Outstanding Achievements in International Economics*	*2024*
International Economics Section Summer Fellowship, Princeton University	*2023 - 2025*
Real Estate Research Institute (RERI) funding	*2022*
Graduate Fellowship, Princeton University	*2021 - 2027*
Social Science Scholarship, The University of Chicago	*2017 - 2019*

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